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9702/31

May/June 2011

2 hours

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

DO NOT WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **9** printed pages and **3** blank pages.

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You may not need to use all of the materials provided.

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- 1 In this experiment you will investigate how the characteristics of a circuit vary with its resistance.

(a) Connect the circuit of Fig. 1.1.

You should expect to spend at least 15 minutes setting up your circuit.

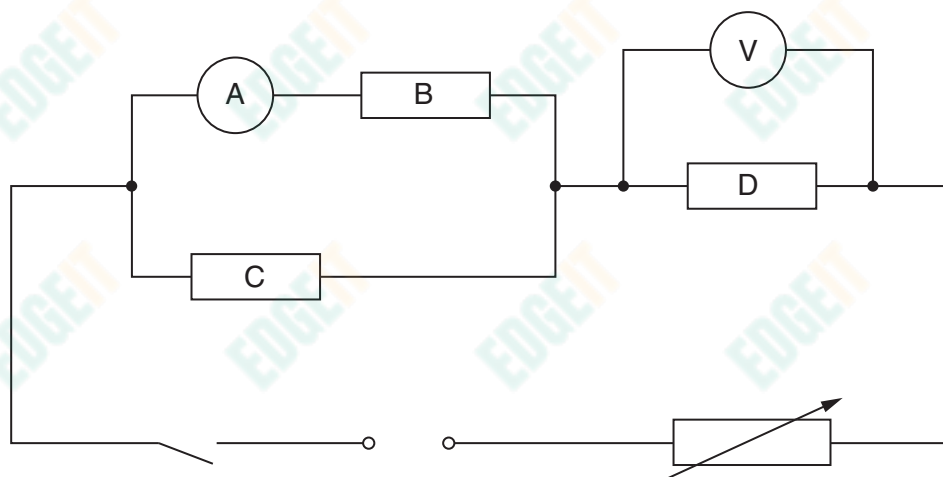


Fig. 1.1

- (b) (i) Close the switch.
 (ii) Adjust the variable resistor until the current reading is at a maximum.
 (iii) Measure and record the ammeter reading I and the voltmeter reading V .

$I =$

$V =$

[2]

- (c) Adjust the variable resistor and repeat (b)(iii) until you have six sets of values of I and V . Include values of $\frac{1}{I}$ and $\frac{1}{V}$ in your table.

Open the switch when you have taken all your readings.

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[10]

- (d) (i) Plot a graph of $\frac{1}{I}$ on the y -axis against $\frac{1}{V}$ on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

[1]

- (iii) Determine the gradient and the y -intercept of this line.

gradient =

y -intercept =

[2]

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- (e) It is suggested that the quantities I and V are related by the equation

$$\frac{1}{I} = \frac{3R}{V} + k$$

where R and k are constants.

Use your answer in (d)(iii) to determine the value of R .

Give appropriate units.

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$R = \dots\dots\dots$ [2]

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- 2 In this experiment you will investigate how the motion of a metre rule balanced on a cylinder depends on the diameter of the cylinder.

(a) Measure and record the thickness t of the metre rule.

$t =$ [1]

(b) (i) Measure and record the diameter d of cylinder A.

$d =$ [1]

(ii) Calculate w , where $w = d - t$.

$w =$ [1]

(c) (i) Use modelling clay to secure cylinder A to the bench and balance the metre rule on the cylinder, as shown in Fig. 2.1.

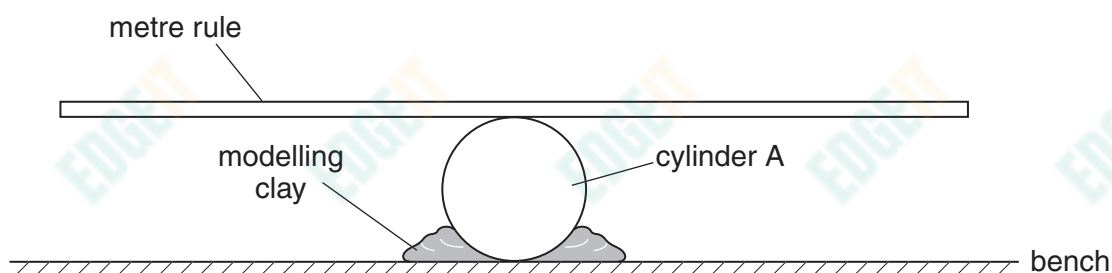


Fig. 2.1

- (ii) Move one end of the rule downwards.
Release the rule and watch the movement.
The end of the rule will move upwards and then downwards again, completing a swing as shown in Fig. 2.2.
The time taken for each complete swing is T .

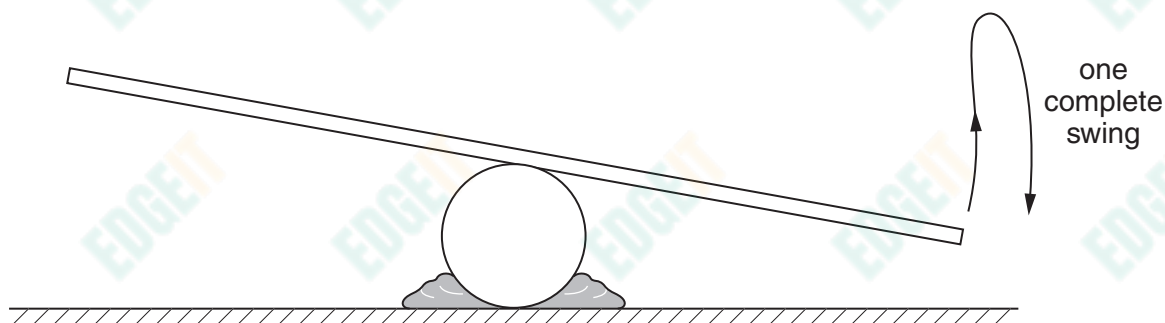


Fig. 2.2

By timing several of these complete swings, determine an accurate value for T .

$T =$ [2]

- (d) Estimate the percentage uncertainty in your value of T .

percentage uncertainty = [1]

- (e) Repeat (b) and (c) for cylinder B.

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$d =$

$w =$

$T =$

[4]

- (f) It is suggested that the quantities T and w are related by the equation

$$T^2 = \frac{k}{w}$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k =$

second value of $k =$

[1]

- (ii) Explain whether your results support the suggested relationship.

.....

 [1]

- (g) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

1.

.....

2.

.....

3.

.....

4.

.....

[4]

- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.

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2.

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3.

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4.

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[4]

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